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RentaCar Management System: Use Case Description

Github: <https://github.com/uyenhoangtp/CS425SE>

1. Introduction

The RentaCar system is a centralized Car Rental Management Web/Desktop Application designed to automate rental processing, track vehicle status in real-time, and provide quick booking tools. It provides a lightweight, straightforward user experience tailored for local rental operations, mitigating the risk of double-bookings and manual data entry errors.

2. Use-Case Model

2.1 Use-Case Diagram Outline

For your PowerPoint presentation, you can draw a simple UML Use-Case diagram with the following relationships:

- **Customer (Actor):** Connects to -> *Search Vehicles*, *Reserve Vehicle*.
- **Admin / Rental Staff (Actor):** Connects to -> *Vehicle CRUD* (Add/Update/Delete), *Manage Vehicle Status*.

2.2 Use-Case Descriptions

Use Case 1: Vehicle CRUD

Feature	Description
Use Case Number	1
Name	Vehicle CRUD
Brief description	This use case allows the Admin and Rental Staff to add, update, or remove vehicles from the database to reflect fleet expansion or changes.
Actors	Admin, Rental Staff.
Preconditions	The Admin or Staff must be logged into the system.

Flows of Events: 1.1.0 Create Vehicle

Step	User Actions	System Actions
1	The user calls the create vehicle command.	The system displays the vehicle entry form requiring Make, Model, Year, Category (SUV, Sedan, Truck), and License Plate.

2	The user fills out the form and requests the system to save the details.	The system verifies there is no existing vehicle with the same License Plate. It saves the vehicle and returns a success message.
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Flows of Events: 1.1.1 Update Vehicle Status

Step	User Actions	System Actions
1	The user requests a list of vehicles.	The system returns a list of all vehicles.
2	The user selects a vehicle to update its status to "Maintenance".	The system displays an editable form for the vehicle.
3	The user changes the status and saves.	The system updates the vehicle status and removes it from the customer view.
Feature		Description
Postconditions		The vehicle details are persisted/updated in the system database.
Business Rules		System must scale easily as new cars are bought or retired. Vehicles in "Maintenance" must not be available for rent.

Use Case 2: Reserve a Vehicle

Feature	Description
Use Case Number	2
Name	Reserve Vehicle
Brief description	This use case allows a Customer to check vehicle availability for a specific timeframe and secure a reservation.
Actors	Customer.
Preconditions	The Customer must be logged into the system using role-based secure credentials.

Flows of Events: 1.2.0 Search and Book Vehicle

Step	User Actions	System Actions
1	The Customer enters desired rental dates and vehicle category to search.	The system filters the database and returns a list of available vehicles matching the criteria.
2	The Customer picks an available vehicle and submits a reservation block.	The system executes automated overlap validation rules to check vehicle availability.
3		The system confirms the reservation and returns a success message, updating the car's schedule. If database operations fail mid-booking, transactions must fully roll back.
Feature		Description
Postconditions		The reservation block is saved, and the vehicle is marked as reserved for those dates.
Business Rules		Overbooking prevention: Two customers cannot book the same car at the same time.